18. When a molecule or ion is reabsorbed from the lumen of the nephron, where does it go? If a solute is filtered and not reabsorbed from the tubule, where does it go?

19. Which segment is the site of secretion and regulated reabsorption of ions and pH homeostasis?

20. What solute is normally present in the body to estimate GFR in humans?

21. Which part of the autonomic nervous system is involved in micturition process?

22. If the afferent arteriole of the nephron constricts, what happens to the GFR in that nephron? If the efferent arteriole constricts what happens to the GFR in that nephron? Assume that no auto regulation takes place.

23. Identfiy the biological term

    Excretion, glomerulus, urinary bladder, glomerular filtrate, ureters, urine, Bowman's capsule, urinary system, reabsorption, micturition, osmosis, proteins, homeostasis.

    a. A liquid which gathers in the bladder.
    b. Produced when blood is filtered in a Bowman's capsule.
    c. Temporary storage of urine.
    d. A ball of inter twined capillaries.
    e. Removal of unwanted substances from the body.
    f. Each contains a glomerulus.
    g. Carry urine from the kidneys to the bladder.
    h. Scientific term for urination.
    i. Regulation of water and dissolved substances in blood and tissue fluid.
    j. Consists of the kidneys, ureters and bladder.
    k. Removal of useful substances from glomerular filtrate.
    l. What solute the blood contains that are not present in the glomerular filtrate?

24. With regards to toxicity and the need for dilution in water, how different are ureotelic and uricotelic excretions? Give examples of animals that use these types of excretion?

25. Differentiate protonephridia from metanephridia.

26. What is the nitrogenous waste produced by amphibian larvae and by the adult animal?

27. How is urea formed in the human body?

28. Differentiate cortical from medullary nephrons.

29. What vessels carry blood to the kidneys? Is this blood arterial or venous?

30. Which vessels drain filtered blood from the kidneys?

31. What is tubular secretion? Name the substances secreted through the renal tubules.

32. How are the kidneys involved in controlling blood volume? How is the volume of blood in the body related to arterial pressure?

33. Name the three main hormones involved in the regulation of the renal function?

34. What is the function of antidiuretic hormone? Where is it produced and what stimuli increases or decreases its secretion?

35. What is the effect of aldosterone on kidneys and where is it produced?

36. Explain the heart's role in secreting a hormone that regulates renal function? What hormone is this?

XI Std Zoology Chapter-8 EM.indd 152 30/12/2022 13:48:09